

sheth l u jhaveri college of arts and sir m v college of science and commerce
Practical no. 8 AI
K-Nearest Neighbors (K-NN)

AIM : Implement the K-NN algorithm for classification or regression.

Apply the K-NN algorithm to a given dataset and predict the class or value for test data.

Evaluate the accuracy or error of the predictions and analyze the results.

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ai p8 T117.py - C:\Users\Adarsh\OneDrive\Videos\ai p8 T117.py (3.12.6)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

print("Adarsh Shukla T117")

df = pd.read_csv(r"C:\Users\Adarsh\Downloads\winemag-data_first150k.csv")

print("\nDataset:")
print(df.head())

df = df[["country", "points", "price"]].dropna()

top_countries = df["country"].value_counts().head(5).index
df = df[df["country"].isin(top_countries)]

X = df[["points", "price"]]
y = df["country"]

encoder = LabelEncoder()
y = encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y)
```

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    X,
    y,
    test_size=0.2,
    random_state=42,
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)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

k_values = range(1, 16)
accuracies = []

for k in k_values:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train_scaled, y_train)
    predictions = model.predict(X_test_scaled)
    accuracies.append(accuracy_score(y_test, predictions))

best_k = list(k_values)[np.argmax(accuracies)]

model = KNeighborsClassifier(n_neighbors=best_k)
model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)

accuracy = accuracy_score(y_test, y_pred)

print("\nBest K:", best_k)

30°C
Partly cloudy
Ln: 100 Col: 0
20:09
05-10-2028

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print("\nBest K:", best_k)
print("Accuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=encoder.classes_,
        zero_division=0
    )
)

plt.figure(figsize=(9, 5))
plt.plot(list(k_values), accuracies, marker="o")
plt.xlabel("K Value")
plt.ylabel("Accuracy")
plt.title("K-NN Accuracy for Different K Values")
plt.xticks(list(k_values))
plt.grid()
plt.show()

sample_size = min(1500, len(X_test))

plt.figure(figsize=(10, 6))

scatter = plt.scatter(
    X_test.iloc[:sample_size]["points"],
    X_test.iloc[:sample_size]["price"],
    c=y_pred[:sample_size],
    alpha=0.7
)
```

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    c=y_pred[:sample_size],
    alpha=0.7
)

plt.xlabel("Wine Points")
plt.ylabel("Wine Price")
plt.title("K-NN Wine Country Classification")
plt.colorbar(scatter, label="Predicted Country Class")
plt.show()

print("\nCountry Classes:")

for i, country in enumerate(encoder.classes_):
    print(i, "=", country)
```

Output :

```
IDLE Shell 3.12.6
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===== RESTART: C:/Users/Adarsh/OneDrive/Videos/ai p8 T117.py =====
Adarsh Shukla T117

Dataset:
  Unnamed: 0 country ... variety winery
0         0   US ... Cabernet Sauvignon Heitz
1         1 Spain ... Tinta de Toro Bodega Carmen Rodriguez
2         2   US ... Sauvignon Blanc Macauley
3         3   US ... Pinot Noir Ponzi
4         4 France ... Provence red blend Domaine de la Bégude

[5 rows x 11 columns]

Best K: 14
Accuracy: 54.59 %

Classification Report:
      precision    recall  f1-score   support

    Chile       0.20       0.10       0.14       1153
    France       0.32       0.12       0.17       2957
    Italy        0.30       0.19       0.23       3757
    Spain        0.20       0.07       0.11       1632
    US           0.62       0.86       0.72      12428

 accuracy         0.55       0.55      21927
  macro avg       0.33       0.27       0.27      21927
 weighted avg     0.47       0.55       0.49      21927

Country Classes:
0 = Chile
```

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```

IDE Shell 3.12.6
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0 0 US ... Cabernet Sauvignon Heitz
1 1 Spain ... Tinta de Toro Bodega Carmen Rodriguez
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[5 rows x 11 columns]

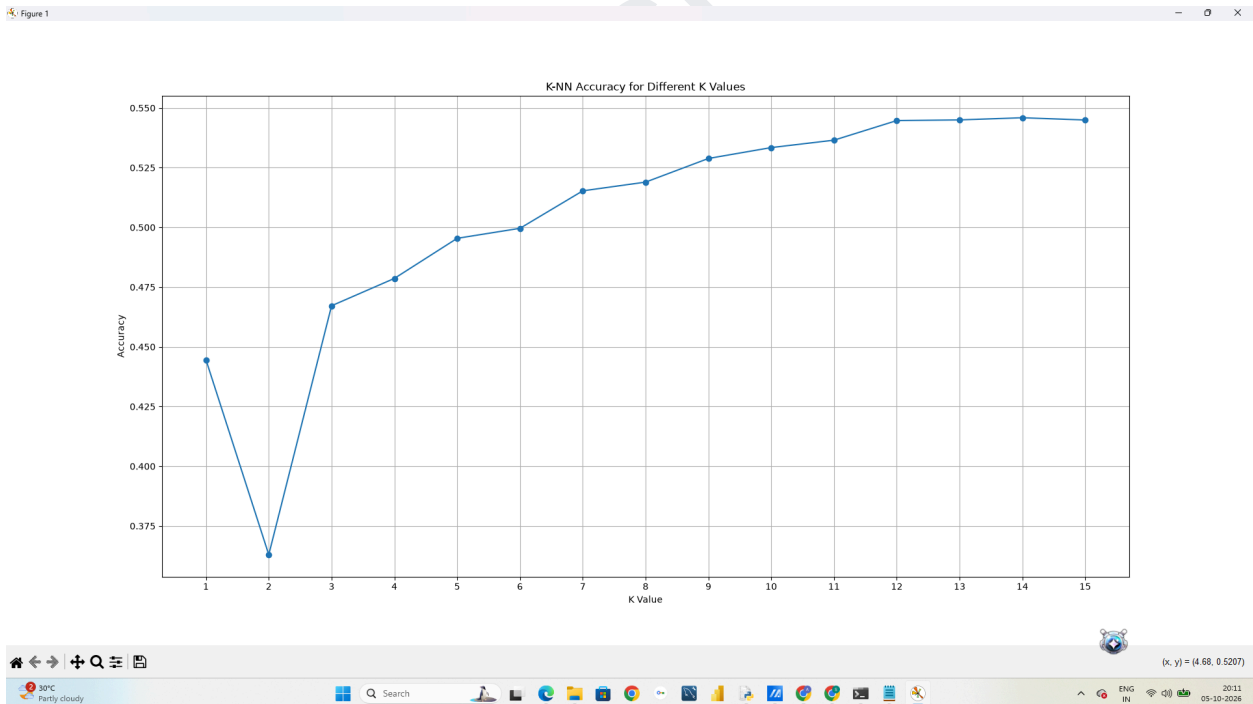
Best K: 14
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    Chile      0.20      0.10      0.14      1153
    France     0.32      0.12      0.17      2957
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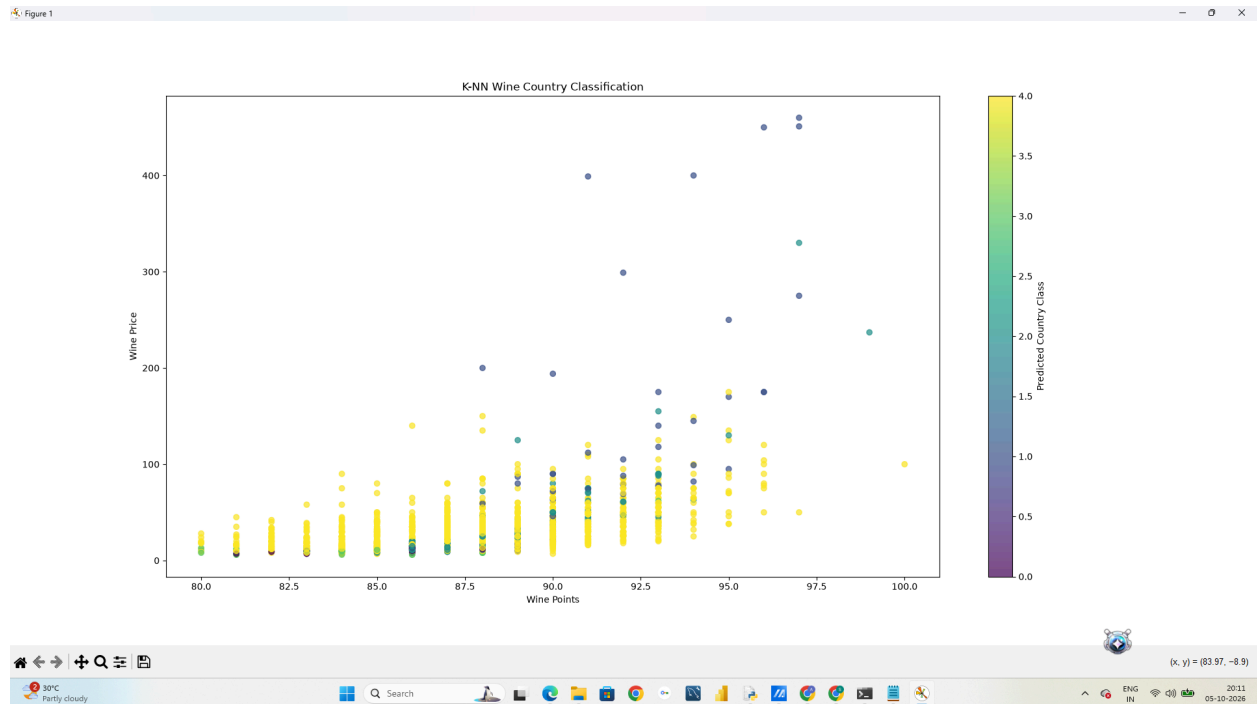
 accuracy      0.55      0.55      0.55     21927
  macro avg    0.33      0.27      0.27     21927
 weighted avg   0.47      0.55      0.49     21927

Country Classes:
0 = Chile
1 = France
2 = Italy
3 = Spain
4 = US
  
```



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